



BEYOND A CHECKER

An Explainable, AI-Supported Advisor
for Radiation Safety Report Assessment



*Helping DSA turn manual review into faster, more consistent, and
more transparent case handling in ServiceNow*

TEAM DEEP THINKER

Presented to:

KPMG & ServiceNow Hackathon 2026

DSA – Norwegian Radiation
and Nuclear Safety Authority

MEET THE TEAM



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EXECUTIVE SUMMARY



1. THE CASE

DSA reviews radiation safety reports manually today. We saw an opportunity to use AI in ServiceNow to assess reports, explain gaps, and support decisions.

- Unstructured Word/PDF reports against DSA Guide 9
- Manual review is slow, variable, and hard to audit
- Explainability and traceability are critical requirements



2. APPROACH & SOLUTION

We approached this as a decision-support problem: identify the issue, shape the solution, build in ServiceNow, and validate through demo.

- AI extracts and maps report content to Guide 9 requirements
- ServiceNow structures, routes, and stores the assessment
- DSA officer reviews, validates, and confirms the decision



3. WAY FORWARD

Strong potential in efficiency, consistency, and explainability, with clear next steps in evidence extraction, reviewer tooling, and reuse.

- Stronger evidence references and report section links
- Automated follow-up tasks and richer reviewer summaries
- Reusable pattern for similar public-sector regulatory cases

WHY THIS MATTERS

At DSA, expert time is valuable. Handlers must manually review long safety reports against DSA Guide 9, compare them to requirements, and document findings in a consistent, explainable, and well-supported way.

THE CHALLENGE

Expert judgment should be spent on decisions, not document handling.

~20%

of work time spent searching for and gathering information

68%

say they lack enough uninterrupted focus time

Doc-heavy

review slows expert work and is harder to standardize



Safety reports are long, document-heavy, and time-consuming to assess



Reports mix structured and unstructured information



Assessments must be consistent, explainable, and well documented



Better tooling improves both productivity and decision quality

THE OPPORTUNITY

ServiceNow can change the game



AI

AI-powered assessment

Interpret report content, map requirement compliance, and generate structured first-pass assessments. The AI does not just flag - it explains.



UNSTRUCTURED DATA

Data standardization

Turn Word/PDF safety reports into structured assessment objects - reusable, reviewable, and fully stored inside ServiceNow.



PROCESS AUTOMATION

Workflow & human-in-the-loop

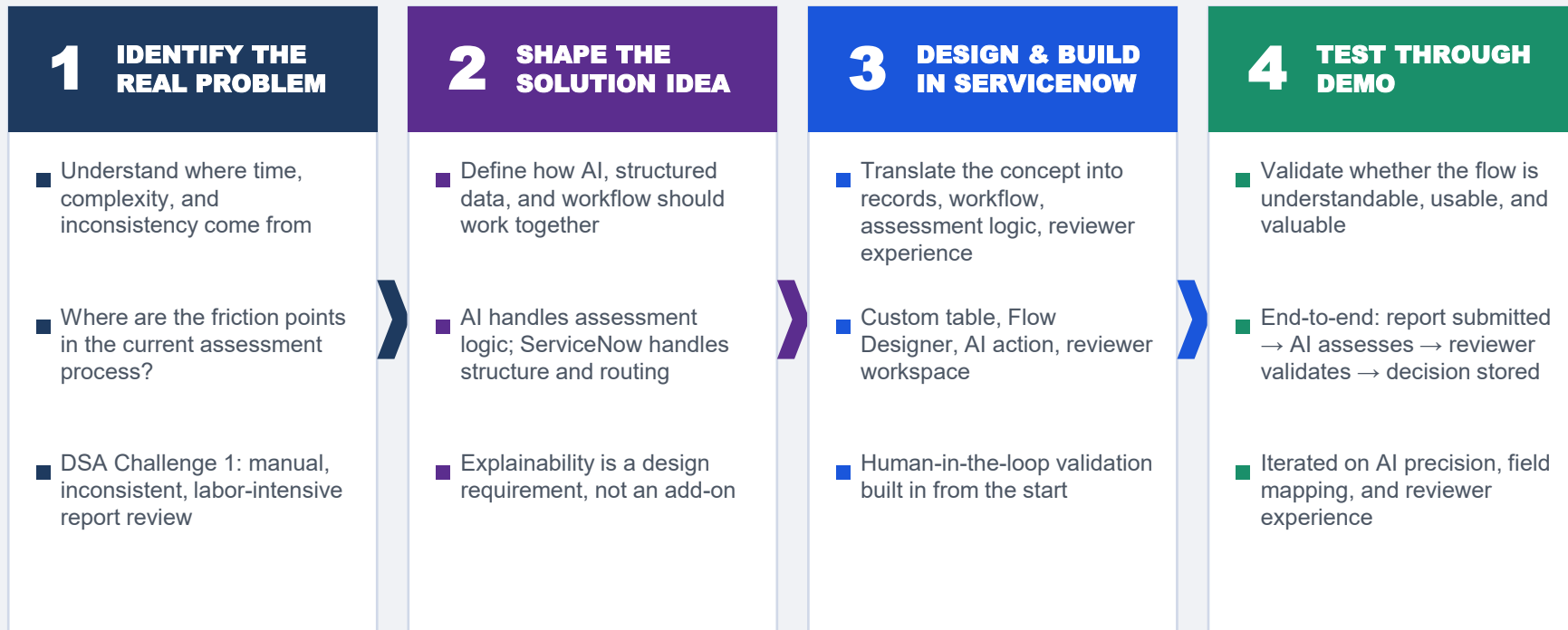
Embed AI assessment into a governed workflow with reviewer validation and clear decision handling in one process.

Why this case: AI × unstructured data × workflow automation — the broadest set of ServiceNow capabilities in one solution.

Illustrative benchmark context, not current DSA performance. Sources: McKinsey (2012); Microsoft Work Trend Index (2023).

HOW WE REALIZED THE OPPORTUNITY

We approached this as a ServiceNow-enabled decision-support problem. Starting from one key question: What would genuinely help a DSA case handler make faster, clearer, and more consistent decisions?



HACKATHON WEEKEND — 20–22 March 2025

FROM CORE PROBLEMS TO OUR SOLUTION

We turned the case's core challenges into three capabilities of an AI advisor.

FROM



Long, unstructured safety reports are hard to assess consistently



Reviewers need reasoning they can trust - the brief explicitly stresses explainability as critical



Analysis only creates value if it fits the real workflow and integrates with case handling

TO

AI-powered requirement assessment

- System reads and parses the full report content automatically
- Maps each section to Guide 9 requirements: fulfilled, partial, or missing
- No manual cross-referencing - the AI does the scanning

Explainable decision support

- Every finding comes with a deviation description and the reason it matters
- Evidence excerpts pulled from the report are shown alongside the conclusion
- Guide 9 clause and report section referenced for full traceability

ServiceNow-native workflow orchestration

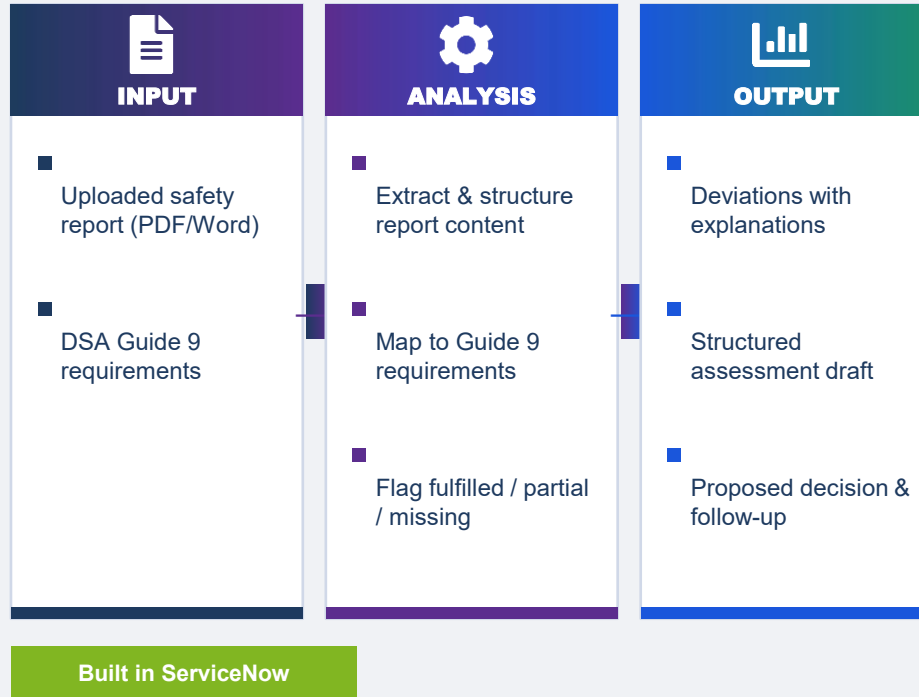
- Assessment and draft decision stored as structured objects in ServiceNow
- Reviewer workspace surfaces findings directly in the case record
- Follow-up tasks triggered automatically when information is missing

HOW THE SOLUTION WORKS IN PRACTICE

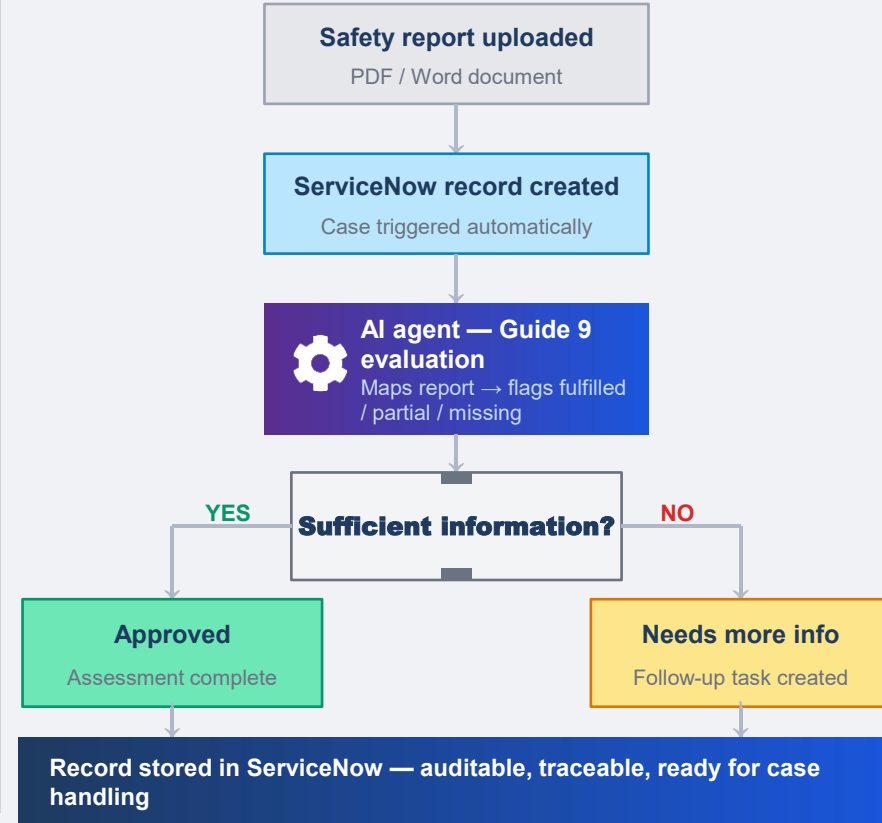
PROCESS / JOURNEY	1 Applicant submits safety report	2 Record created in ServiceNow	3 AI analyzes report content	4 Assessment & proposed decision generated	5 Reviewer validates and adjusts	6 Final outcome stored for case handling
SERVICENOW COMPONENTS	Custom table & data model	Flow Designer / Workflow Studio	AI step / AI action	Requirement fields + assessment object	Reviewer decision & override fields	Case visibility & stored record
BUILD INSIGHTS	Standardizing unstructured report ingestion	Mapping AI output to structured SN fields	Making reviewer correction loops intuitive	Strengthening evidence references	Enabling reuse for similar regulatory cases	Scaling to more report types & templates

HOW IT WORKS BEHIND THE SCENCE

TECHNICAL FLOW



DECISION FLOW



DEMO

KEY POINT

01

Case overview

02

Requirement-level
assessment

03

Explanation and
references

04

Reviewer validation

05

Proposed decision &
follow-up

[[LINK](#)]

BUSINESS VALUE

*Expected impact based on
enterprise AI & automation benchmarks*

Four measurable outcomes from our ServiceNow AI solution, grounded in industry-standard GenAI benchmarks.



AI EXTRACTION & MATCHING

60–80%

reduction in initial review time

AI-supported extraction and requirement matching

Instead of 2 hours reading a 50-page PDF, the AI highlights exact clauses and maps them to Guide 9 in seconds. The expert spends 30 minutes verifying — not hunting.

GenAI document analysis benchmarks



STRUCTURED REVIEW

100%

assessment traceability — zero omissions

Structured requirement-by-requirement review

Moving from free-text notes to a system-guided ServiceNow checklist mathematically eliminates the risk of skipping a regulatory requirement due to fatigue.

Process compliance & structured workflow standards



DRAFT ASSESSMENT

50–70%

faster documentation

Automatic draft assessment with explanations

If typing a deviation report normally takes 45 minutes, an AI-generated draft reduces that to 10–15 minutes of reviewing and editing. Consistent with Now Assist benchmarks.

ServiceNow Now Assist GenAI benchmarks



DECISION & FOLLOW-UP

100%

elimination of process white space

Proposed decision and automated follow-up

White space — dead time between decision and action — is eliminated. The ServiceNow workflow triggers follow-up instantly, cutting end-to-end case cycle time drastically.

Workflow automation & BPM industry benchmarks

ROADMAP

NOW	SHORT TERM	LONG TERM
Working prototype ■ AI requirement-level assessment ■ Structured reasoning and deviation explanation ■ Reviewer-supported decision handling in ServiceNow ■ Human-in-the-loop validation demonstrated end-to-end	Improve & strengthen ■ Stronger evidence extraction and report references ■ Richer reviewer summary and case-summary views ■ Reviewer correction loop improvements ■ Automated follow-up tasks for missing information	Scale & reuse ■ Dashboards and analytics across multiple cases ■ Broader report/template coverage ■ Reusable patterns for similar regulatory cases ■ Roadmap extension to other DSA assessment types

THANK YOU

Team Deep Thinker